a Validation targets support different claims

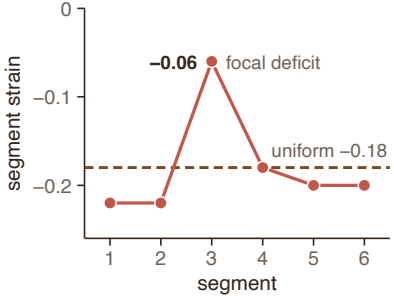
● Validation target	✓ Direct claim it supports	✕ Does not establish on its own
Known motion phantom, STRAUS, MRXCAT2.0	technical error in the simulated regime; capacity needs direct field fitting	in-vivo accuracy
Paired DENSE or tagging same session, registered	material-sensitive agreement under matched definitions	focal location or extent, unless measured
Scan–rescan same subject, re-imaged	precision; accuracy needs a reference	bias
LGE or tissue characterisation scar or fibrosis label	biological association; motion error stays unmeasured	pointwise displacement error
Outcome events, prognosis	prognostic association; utility needs a decision study	decision impact

b Proposed programme: combine complementary tests

Stage	What is varied or fixed	Resources (Table 4)
Pre-specify measurement contract	estimand, reference, tolerance and data splits fixed in advance	
Stage 1 Known motion	vary width, extent, amplitude, timing, resolution and noise	STRAUS **3 templates × 6 states** MRXCAT2.0 **programmable generator** CMAC **dynamic phantom**
Stage 2 Paired material reference	cine with DENSE or tagging in one session; registered; reference blinded	Stanford **55; pairs unverified** CMAC **15 volunteers, 12 landmarks**
Stage 3 Tissue and decisions	LGE, responsiveness, decision impact	Clinical cohorts **LGE or outcomes**

Counts are resource units, not independent test participants.

c Same mean, focal deficit



segments 1–3, focal	**−0.22 · −0.22 · −0.06**
segments 4–6, focal	**−0.18 · −0.20 · −0.20**
global mean, both fields	**−0.18**
range, focal / uniform	**0.16 / 0**

Recovering the mean does not establish deficit recovery.